

Databases for data analytics

<https://github.com/evidencebp/databases-course/>

Window Functions

Window Functions Goal

- Usually we look at a single raw record
- Grouping allows us to look on groups of records
- Window functions allows us to look at a single record and “look through the window” on the context

Window definition

- Current record (implicit yet very important)
- Name (in order to reuse it)
- Partition (the relevant grouping)
- Order (sorting the window)
- Frame (context filtering)

Window is a (rather) good metaphor

- Current record - floor
- Name (e.g., left, center, right)
- Partition (**bad metaphor**
 , maybe sides)
- Order (going up or down)
- Frame (which neighboring
 floors can we see)

Given a floor,
specific window functions compute
a value on the floor.



Common uses

Usually applied to aggregated data in BI systems

- Looking at a month before/after
- Rolling metric (e.g., last 3 month average)
- Specific value (instead of max, the 10th value)
- Percentiles
- Relative position

Resources

- [Window functions](#)
- Sales [examples](#)
- See IMDB [examples](#)

Windows functions implementation has sharp edges and database differences (e.g., count distinct, relations to group by). It is out of the course scope yet remember to use carefully.